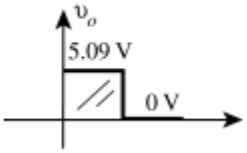
Model answer of sheet 4

1

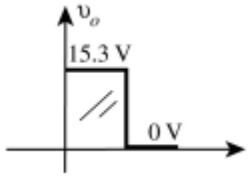
(a) Positive pulse of v_i:

$$V_o = \frac{1.8\ \text{k}\Omega(12\ \text{V} - 0.7\ \text{V})}{1.8\ \text{k}\Omega + 2.2\ \text{k}\Omega} = \mathbf{5.09\ V}$$

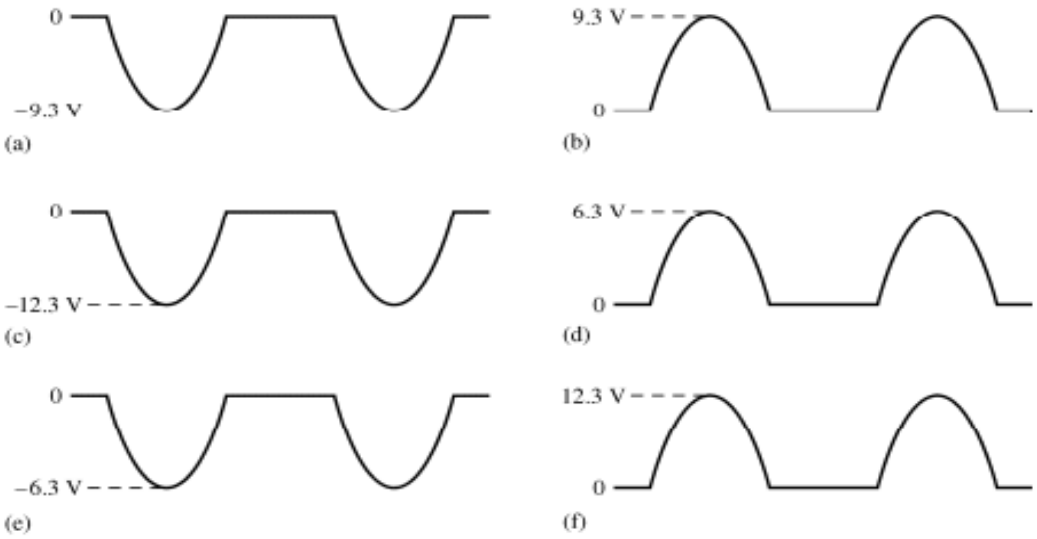
Negative pulse of v_i:
diode "open", $v_o = \mathbf{0\ V}$



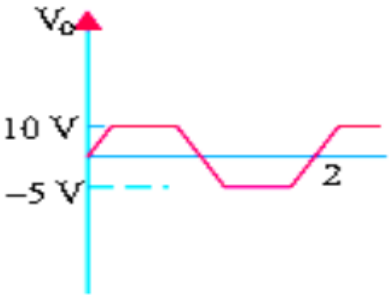
(b) Positive pulse of v_i:
$V_o = 12\ \text{V} - 0.7\ \text{V} + 4\ \text{V} = \mathbf{15.3\ V}$
Negative pulse of v_i:
diode "open", $v_o = \mathbf{0\ V}$



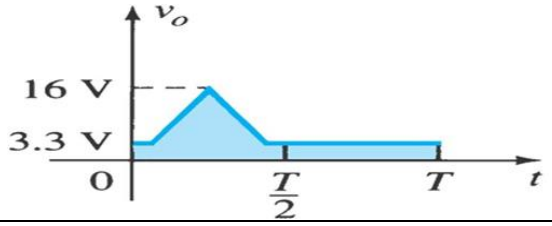
2

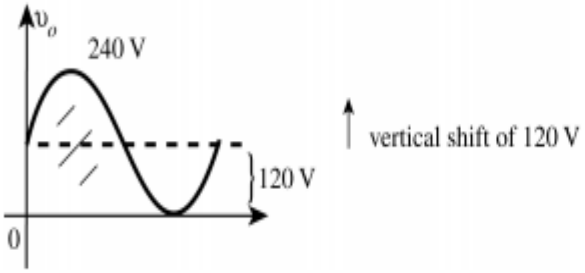
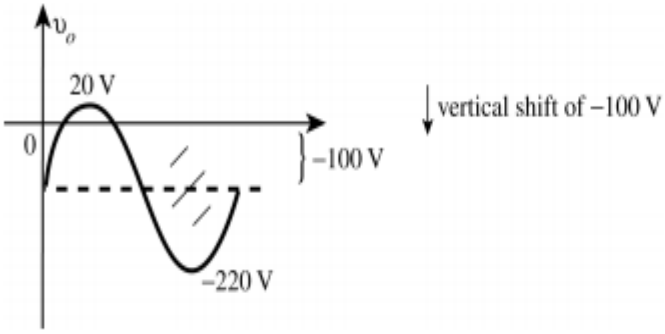
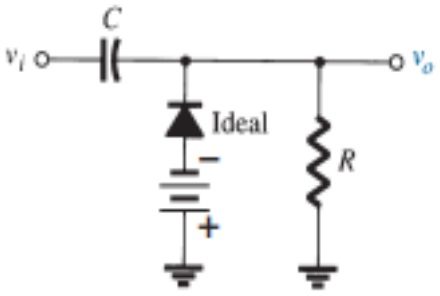


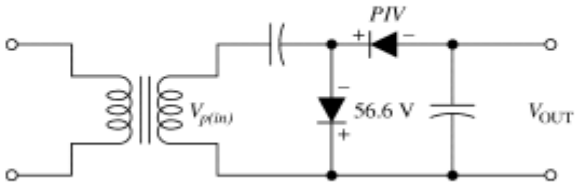
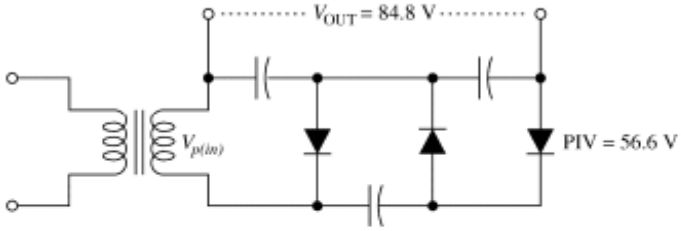
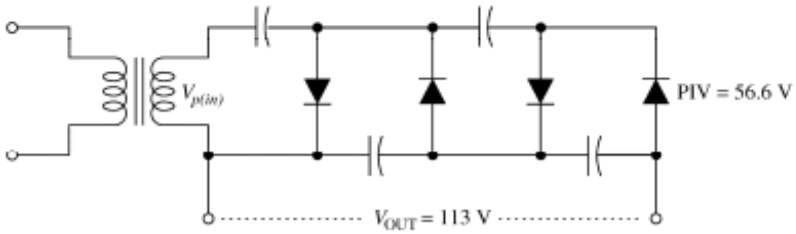
3



4



5	(a) A sine wave with a positive peak at 0.7 V, a negative peak at −7.3 V, and a dc value of −3.3 V. (b) A sine wave with a positive peak at 29.3 V, a negative peak at −0.7 V, and a dc value of +14.3 V. (c) A square wave varying from +0.7 V to −15.3 V with a dc value of −7.3 V. (d) A square wave varying from +1.3 V to −0.7 V with a dc value of +0.3 V.
6	(a) For negative half cycle capacitor charges to peak value of 120 V = 120 V with polarity (− ⊣⊢ +). The output v_o is directly across the "on" diode resulting in v_o = **0 V** as a negative peak value. For next positive half cycle $v_o = v_i + 120$ V with peak value of v_o = 120 V + 120 V = **240 V**.  (b) For positive half cycle capacitor charges to peak value of 120 V − 20 V = 100 V with polarity (+ ⊣⊢ −). The output v_o = 20 V = **20 V** For next negative half cycle $v_o = v_i$ − 100 V with negative peak value of v_o = −120 V − 100 V = **−220 V**. 
7	

8	$V_{OUT} = 2V_{p(in)} = 2(1.414)(20\text{ V}) = \mathbf{56.6\text{ V}}$ See Figure 2-12.  $V_{OUT(trip)} = 3V_{p(in)} = 3(1.414)(20\text{ V}) = \mathbf{84.8\text{ V}}$ $V_{OUT(quad)} = 4V_{p(in)} = 4(1.414)(20\text{ V}) = \mathbf{113\text{ V}}$ See Figure 2-13.  (a) Tripler  (b) Quadrupler